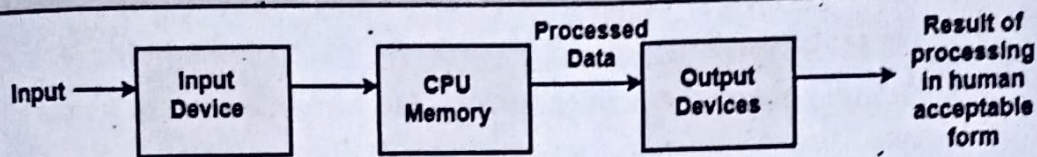


GRAPHICS HARDWARE

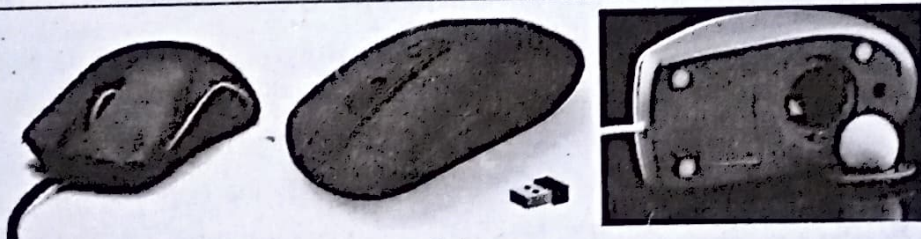
- 2.1 Interactive Input Devices
- 2.2 Display Devices and Hard Copy Devices
- 2.3 Raster and Random Systems and Architectures
- 2.4 Video Controller
- 2.5 Use of Digital to Analog Converter and Frame Buffer Organization
- 2.6 Color Monitors

2.1 Interactive Input Devices



The hardware that is utilized to convey input to the computer is known as the Input Devices. The information can take the form of text, images, sound, and text. Data from the computer's memory is displayed through an output device. Text, numerical data, lines, polygons, and other objects can all be included in the output.

1. Mouse



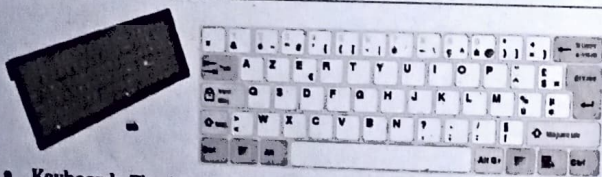
- The mouse is an input device commonly used for computer graphics and general computer usage.
- It allows users to control the position of the cursor on the screen by moving the mouse across a surface.
- The mouse typically has two primary buttons, left and right, which are used for selecting, clicking, and performing various actions.
- A scroll wheel is often located between the buttons, enabling users to scroll vertically through documents, web pages, or other content.
- Mouse clicks can trigger different actions, such as selecting objects, opening context menus, or initiating commands.

- Mice come in different types, including wired and wireless options, with various additional features like programmable buttons or customizable sensitivity settings.

Types

- Optical Mouse:** Uses an LED or laser to track movement by capturing images of the surface beneath the mouse. Affordable and works on various surfaces.
- Laser Mouse:** Utilizes laser technology for precise tracking and sensitivity. Suitable for tasks requiring fine cursor control and can work on different surfaces.
- Wireless Mouse:** Offers freedom of movement without a physical cable by using wireless technologies like Bluetooth or USB receivers. Convenient and ideal for laptops and presentations.
- Gaming Mouse:** Specifically designed for gamers with features like programmable buttons, customizable sensitivity, and ergonomic designs for enhanced gaming performance.
- Trackball Mouse:** Features a stationary ball on top that users manipulate to control the cursor. Provides precision, and comfort, and reduces physical movement.
- Vertical Mouse:** Ergonomically designed with a vertical orientation to reduce strain and promote a more natural hand position. Helps alleviate wrist and hand discomfort.
- Gaming Mousepad with Built-in Mouse Sensors:** Specialized mousepad with integrated sensors to provide enhanced precision and accuracy for gaming or graphic design tasks. Allows for smoother cursor control.
- Mechanical Mouse:** Mechanical mice use a physical ball and internal mechanical mechanisms to track movement. As the user moves the mouse, the ball rotates, and the internal mechanisms detect the rotation and translate it into cursor movement.

2. Keyboard



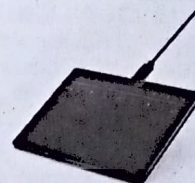
- **Keyboard:** The keyboard is a primary input device used for entering text, commands, and interacting with computer

systems. It consists of a set of keys arranged in a specific layout, such as QWERTY or AZERTY, and provides a tactile and efficient means of input.

- **Key Layout:** Keyboards typically feature alphanumeric keys, function keys, modifier keys (e.g., Shift, Ctrl, Alt), navigation keys (e.g., arrow keys), and various specialized keys (e.g., Enter, Backspace, Delete). The arrangement and labeling of keys follow standard conventions, allowing users to type and execute commands accurately.
- **Typing Efficiency:** Keyboards are designed for touch typing, where users can input text without looking at the keys. This method relies on key placement and the tactile feedback of key presses. Experienced typists can achieve high typing speeds and accuracy with practice.
- **Keyboard Shortcuts:** Keyboards offer a range of shortcuts that allow users to perform actions or trigger commands quickly. These shortcuts often involve combining modifier keys (e.g., Ctrl, Shift) with other keys, enabling efficient navigation, text formatting, and application-specific functions.
- **Customization:** Many keyboards allow customization options, including programmable keys, macros, and lighting effects. Users can tailor the keyboard's functionality and appearance to suit their preferences or specific use cases, enhancing productivity and personalization.
- **Numeric Keys:** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- **Alphabetic keys:** a to z (lower case), A to Z (upper case)
- **Special Control keys:** Ctrl, Shift, Alt
- **Special Symbol Keys:** ; , " ? @ ~ ? :
- **Cursor Control Keys:** ↑ → ← ↓
- **Function Keys:** F1 F2 F3....F9.
- **Numeric Keyboard:** It is on the right-hand side of the keyboard and used for fast entry of numeric data.

3. Trackpad or Touchpad

- **Trackpad:** Also known as a touchpad, it is an input device commonly found on laptops and some desktop peripherals.
- **Touch Sensitivity:** Detects and responds to finger movements and gestures on its surface.



- **Cursor Control:** Allows users to control the position of the cursor on the screen by moving their fingers on the trackpad.
- **Gestures:** Supports various multi-finger gestures like scrolling, zooming, swiping, and rotating, enhancing navigation and interaction.
- **Button Integration:** Some trackpads have integrated buttons for left-click and right-click actions, eliminating the need for a separate mouse.
- **Portable and Space-saving:** Ideal for laptops and compact setups, as it eliminates the need for external mice, saving desk space and providing a portable input solution.

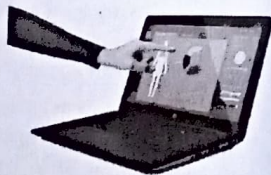
4. Joystick

- **Joystick:** A handheld input device consisting of a lever or stick that can be tilted or moved in different directions to control the movement of an object, typically in video games or flight simulations.
- **Directional Control:** Joysticks provide precise directional control, allowing users to navigate in multiple axes and execute complex movements.
- **Buttons and Triggers:** Joysticks often feature buttons and triggers that can be pressed or activated to perform actions, such as shooting, jumping, or activating special abilities.
- **Immersive Gaming Experience:** Joysticks enhance the gaming experience by providing a more intuitive and engaging way to control characters or vehicles, especially in games that involve flying, driving, or combat.
- **Specialized Joysticks:** There are specialized joysticks designed for specific applications, such as flight simulation or arcade gaming, featuring additional controls, adjustable resistance, or force feedback to simulate realistic sensations and improve immersion.



5. Touch Screen

- **Touch Screen:** A display technology that allows users to interact directly with the screen by touching or tapping on it. It eliminates the need for external input devices like keyboards or mice.



- **Intuitive Interaction:** Touch screens offer a natural and intuitive way of interacting with digital content, making it easy for users to navigate, select, and manipulate on-screen elements.
- **Multi-Touch Support:** Many touch screens are capable of recognizing multiple touch points simultaneously, enabling gestures like pinch-to-zoom, two-finger scrolling, or rotating images.
- **Widely Used:** Touch screens have become ubiquitous in smartphones, tablets, and other handheld devices, enhancing user experience and enabling seamless interaction with apps and multimedia.
- **Resistive vs. Capacitive:** There are two primary types of touch screens: resistive and capacitive. Resistive screens respond to pressure and can be used with stylus or gloved hands, while capacitive screens detect electrical charges from the user's fingertips.
- **Advancements:** Touchscreen technology continues to evolve with advancements like haptic feedback, in-display fingerprint sensors, and edge-to-edge displays, offering more immersive and feature-rich user experiences.

6. Digitizer

- **Digitizer:** A component integrated with a touch screen that converts touch inputs into digital signals, allowing for precise tracking and accurate interpretation of touch gestures.
- **Pen Input:** Digitizers often support pen input, allowing for more precise and pressure-sensitive interactions. This is particularly useful in tasks like digital art, note-taking, or graphic design.
- **Pressure Sensitivity:** Some digitizers are pressure-sensitive, meaning they can detect varying levels of pressure applied by the pen or stylus. This feature enables artists and designers to create detailed and nuanced drawings or paintings.
- **Handwriting Recognition:** Digitizers with handwriting recognition capabilities can convert handwritten inputs into digital text, making them useful for taking digital notes or annotating documents.



- **Active vs. Passive:** Digitizers can be categorized as active or passive. Active digitizers require a powered stylus or pen, while passive digitizers work with any conductive object, like a finger or stylus.
- **Professional Applications:** Digitizers find applications in professional environments like graphic design, CAD (Computer-Aided Design), and medical fields where precise input and accurate tracking are critical for tasks such as drawing diagrams or annotating medical images.

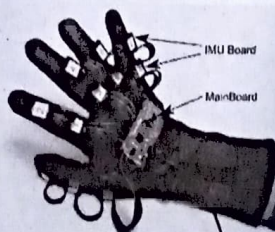
7. Light Pen

- **Light Pen:** A pointing device that uses light-sensitive technology to detect and interact with a computer screen by directly touching or hovering over specific areas.
- **Light Detection:** Light pens work by detecting changes in light intensity on the screen caused by the pen's interaction, allowing for precise input and interaction with graphical elements.
- **Drawing and Selection:** Light pens are often used for drawing, selecting, and editing objects on the screen, providing a natural and precise input method for graphic design or digital art applications.
- **Limited Compatibility:** Light pens require screens with specific properties, such as a cathode ray tube (CRT) display, as they rely on the scanning electron beam used in CRT technology to function effectively.
- **Historical Significance:** Light pens were popular in the early days of computing but have been largely replaced by other input devices, such as touch screens or stylus tablets, which offer broader compatibility and functionality.



8. Data Glove

- **Data Glove:** A wearable input device that resembles a glove and is used to track and interpret hand and finger movements for controlling virtual objects or interacting with computer systems.



- **Motion Tracking:** Data gloves utilize sensors and tracking technology to capture the position and orientation of the wearer's hand and fingers, allowing for precise and intuitive control in virtual or augmented reality environments.
- **Gesture Recognition:** By analyzing the movements of individual fingers and hand gestures, data gloves enable the recognition of complex gestures, providing a natural and immersive way to interact with digital content or control virtual avatars.
- **Haptic Feedback:** Some advanced data gloves incorporate haptic feedback technology, providing tactile sensations or vibrations to the wearer's hand, enhancing the sense of touch and realism in virtual environments.
- **Applications:** Data gloves find applications in fields such as virtual reality gaming, motion capture, training simulations, robotics, and rehabilitation, where precise hand and finger tracking and control are crucial for realistic and intuitive interaction.

9. Scanner

- **Scanner:** A device that converts physical documents, images, or objects into digital format by capturing and digitizing their contents.
- **Optical Scanning:** Scanners use optical sensors or cameras to scan the document or object, capturing its details and converting them into a digital image or text.
- **Flatbed or Sheet-fed:** Scanners come in different forms, such as flatbed scanners that have a glass surface where documents are placed, or sheet-fed scanners that can automatically feed multiple pages.
- **Resolution and Color Depth:** Scanners offer different resolution options, determining the level of detail captured in the scanned image. They also support color depth settings to accurately represent the colors in the original document or image.
- **Document Archiving and Image Editing:** Scanners are commonly used for archiving physical documents, creating digital copies, and enabling easy document management. They also provide a convenient way to scan photos, artwork, or other visual content for editing or sharing digitally.



10. Voice System

- **Voice System:** A technology that enables interaction between users and computer systems through voice commands and responses.
- **Speech Recognition:** Voice systems employ speech recognition algorithms to convert spoken words or commands into text or computer-readable input.
- **Natural Language Processing:** Advanced voice systems utilize natural language processing techniques to understand and interpret the meaning and context of spoken language, allowing for more sophisticated interactions.
- **Voice Response:** Voice systems can provide spoken responses or feedback to users, either through pre-recorded audio or generated speech, enhancing the user experience and enabling hands-free operation.
- **Applications:** Voice systems find applications in various domains, including virtual assistants, voice-controlled smart devices, customer support systems, dictation software, and voice-enabled navigation systems, offering convenient and intuitive ways to interact with technology.



2.2 Hard Copy Devices and Display Devices

Hard copy devices

Printer

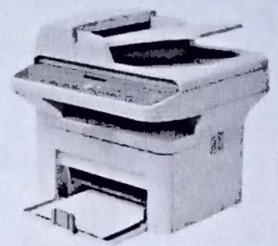
- **Printer:** Device that produces hard copies of digital documents or images on paper or other media using various printing technologies like inkjet or laser.
- **Printing Options:** Printers offer options such as color or black and white printing, different paper sizes, and quality settings to suit specific printing needs.
- **Versatility:** Printers can handle a variety of tasks, including printing documents, photos, labels, and even 3D objects, depending on the printer type.



Photocopier

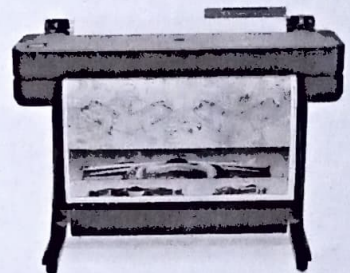
- **Connectivity:** Many printers have built-in wireless or network connectivity, enabling printing from multiple devices or over a network.
- **Home and Office Use:** Printers are widely used in both home and office environments for tasks ranging from everyday document printing to high-quality photo prints or professional documents.

- **Photocopier:** Machine that duplicates physical documents or images by using light and electrostatic processes.
- **Copying Features:** Photocopiers offer options such as resizing, collating, or duplex copying to meet specific copying requirements.
- **Speed and Efficiency:** Photocopiers are designed for quick and efficient copying, often with high-speed capabilities and automatic document feeders for multiple-page copying.
- **Document Enlargement or Reduction:** Photocopiers can resize documents, allowing for enlargements or reductions in size to fit specific requirements.
- **Office Productivity:** Photocopiers are commonly found in office settings, providing convenient and reliable copying services for documents, reports, or other materials.



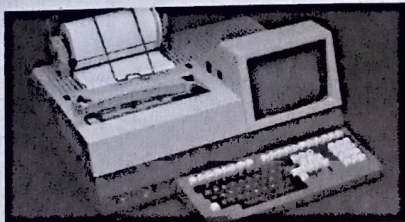
Plotter

- **Plotter:** Specialized output device used for large-format printing, commonly used for architectural or engineering drawings, graphical designs, or technical illustrations.
- **Precision and Detail:** Plotters are designed to produce highly detailed and precise prints, often capable of creating large-scale prints with fine lines, intricate patterns, and accurate colors.
- **Pen or Inkjet Technology:** Plotters utilize either pen-based technology, where pens move across the paper to create the image, or inkjet technology that sprays ink onto the paper.



- **Wide-Format Printing:** Plotters can handle large paper sizes and create prints with widths ranging from 24 inches to several feet, allowing for the production of posters, banners, or large-scale technical drawings.
- **Professional Applications:** Plotters find applications in fields such as architecture, engineering, graphic design, or any industry that requires high-quality, large-format prints with precise details.

Fax



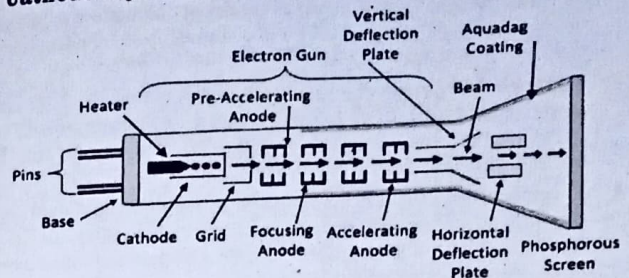
- **Fax:** Communication device that transmits scanned documents over telephone lines to a receiving fax machine, generating physical copies of the sent documents.
- **Document Transmission:** Faxes send documents in a digital format over analog phone lines, allowing for the transmission of text, images, or a combination of both.
- **Real-Time Communication:** Faxes provide real-time document transmission, enabling immediate receipt and printing of documents on the receiving end.
- **Simplified Workflow:** Faxes eliminate the need for physical document transportation or mailing, streamlining communication and document sharing processes.
- **Decreased Reliance:** With the advent of digital communication and email, fax usage has declined, but it remains relevant in certain industries or situations that still rely on fax-based communication or require physical copies of documents.

Types of display devices

There are some display devices given below:

- | | |
|---------------------------------|-------------------------------|
| 1. Cathode-Ray Tube (CRT) | 2. Color CRT Monitor |
| 3. Flat panel display | 4. Plasma display |
| 5. Liquid crystal display (LCD) | 6. Light Emitting Diode (LED) |

1. Cathode ray Tube



Main components

1. **Electron Gun:** An electron gun is made up of a number of components, principally a cathode and a heating filament (warmer). The electron cannon produce an electron source that is narrowed down into a beam and pointed towards the CRT's face.
2. **Control Electrode:** This device is used to activate and deactivate the electron beam.
3. **Focusing system:** It directs the electrons into a narrow beam to produce a clean image.
4. **Deflection yoke:** The direction of electron beams can be adjusted using it. As the electron beam travels across the region, it is bent by a resulting electric or magnetic field. A sweep or scan generator is connected to the yoke of a traditional CRT. A fluctuating magnetic or electric potential is produced by the deflection yoke, which is coupled to the sweep generator.
5. **Phosphorus-coated screen:** The inside front surface of every CRT is covered in phosphors. When an intense electron beam strikes a phosphor, it emits light. The light emitted by a phosphor upon exposure to an electron beam is known as phosphorescence.

Working Mechanism of CRT

- **Electron Generation:** Inside the CRT, a heated cathode emits electrons through thermionic emission. The cathode is typically made of a material like tungsten with high electron emissivity.
- **Electron Acceleration:** The emitted electrons are accelerated by a positively charged anode, which provides an electric field to accelerate the electrons toward the screen.

- **Electron Focusing:** The accelerated electrons pass through a focusing system that consists of focusing electrodes or magnetic fields. This system ensures that the electron beam remains tightly focused and narrow.
- **Electron Deflection:** The focused electron beam is deflected by electromagnetic coils or plates known as deflection yokes. These yokes create magnetic or electric fields to steer the beam horizontally and vertically across the screen.
- **Phosphor Excitation:** As the deflected electron beam strikes the phosphor-coated screen, it causes the phosphor to emit light. The phosphor is a material that converts the kinetic energy of the electrons into visible light.
- **Aquadag Coating:** Inside the CRT, the interior surface is coated with a conductive graphite material called aquadag. Aquadag helps maintain a uniform electrical potential across the screen by reducing electrostatic interference and minimizing the buildup of electrical charges. It is connected to the cathode and acts as a ground reference for the electron beam.
- **Image Formation:** The deflected and focused electron beam scans the phosphor-coated screen in a pattern known as a raster. Varying the intensity of the electron beam as it scans different areas of the phosphor coating creates the desired image or display.
- **Persistence of Vision:** Due to the persistence of vision, the image on the screen appears continuous to the human eye. The phosphor glow lingers for a short time after the electron beam has moved to the next position, creating the illusion of a stable image.

Types of CRT devices

- Direct view storage tube
- Raster scan device
- Random scan device

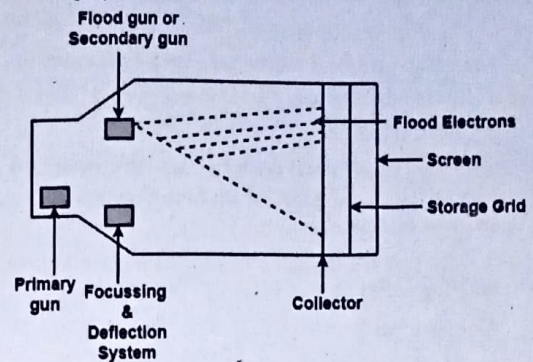
a. Direct view storage Tube

DVST terminals also use the random scan approach to generate the image on the CRT screen. The term "storage tube" refers to the ability of the screen to retain the image which has been projected against it, thus avoiding the need to rewrite the image constantly.

Function of guns: Two guns are used in DVST

1. **Primary guns:** It is used to store the picture pattern.

2. **Flood gun or Secondary gun:** It is used to maintain picture display.



Direct View Storage Tube

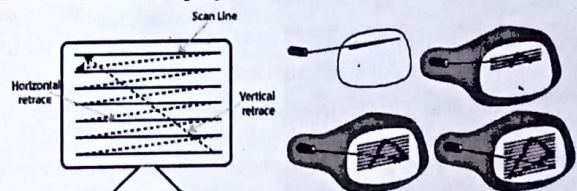
Advantage:

1. No refreshing is needed.
2. High Resolution
3. Cost is very less

Disadvantage:

1. It is not possible to erase the selected part of a picture.
2. It is not suitable for dynamic graphics applications.
3. If a part of picture is to modify, then time is consumed.

b. Raster Scan display



There are two ways to represent an object on the screen:

1. **Raster Scan:** It is a scanning technique in which the electron beam moves along the screen. It moves from top to bottom, covering one line at a time.

A raster scan is based on pixel intensity control display as a rectangular box on the screen called a **raster**.

Picture description is stored in the memory area called as **Refresh buffer, or Frame Buffer**.

Frame buffer is also known as **Raster or Bitmap**. Raster scan provides the refresh rate of 60 to 80 frames per second.

For Example: Television

The beam refreshing has two types:

1. Horizontal Retracing
2. Vertical Retracing

When the beam starts from the top left corner and reaches bottom right, and again return to the top left, it is called the **vertical retrace**.

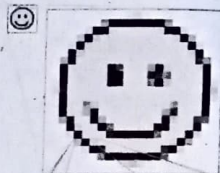
It will call back from top to bottom more horizontally as a horizontal reversal.

Advantages:

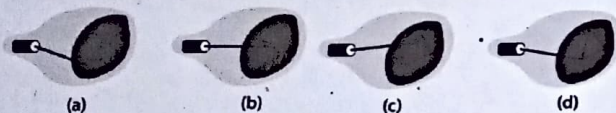
1. Real image
2. Many colors to be produced
3. Dark scenes can be pictured

Disadvantages:

1. Less resolution
2. Display picture line by line
3. More costly
4. Due to scan conversion jaggies or stair-casing is formed.



C. Random Scan display



Random Scan (Vector scan): It is also known as stroke-writing display or calligraphic display. In this, the electron beam points only to the area in which the picture is to be drawn.

It uses an electron beam like a pencil to make a line image on the screen. The image is constructed from a sequence of straight-line segments. On the screen, each line segment is drawn by the beam to pass from one point on the screen to the other, where its x & y coordinates define each point.

After compilation of picture drawing, the system cycle back to the first line and create all the lines of picture 30 to 60 times per second.

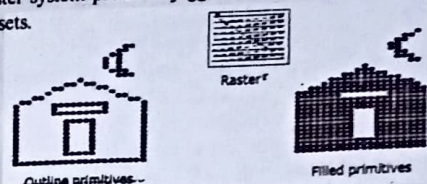
Advantages:

1. High Resolution
2. Draw smooth line Drawing

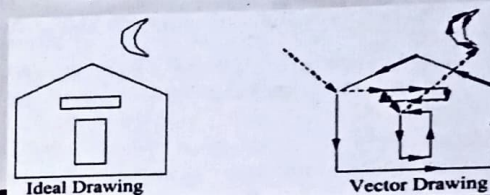
Disadvantages:

1. It does only the wireframe.
2. It creates complex scenes due to flicker.

➤ A Raster system produces jagged lines that are plotted as discrete points sets.



➤ Vector displays produce smooth line drawing



2. Color CRT

A CRT monitor displays color pictures by using a combination of phosphors that emits different colored light. We also have basic phosphors, which can produce three basic colors i.e. Red, Blue and Green. So, one should have a technology to combine them in different combinations.

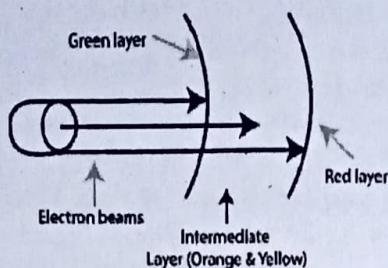
The basic principle behind colored displays is that combining these 3 basic colors - Red, Blue and Green, can produce every color.

By choosing different ratios of these three colors we can produce different colors - millions of them in fact.

There are two popular techniques for producing color displays with a CRT. They are:

- a. Beam Penetration method
- b. The Shadow - Mask method

a. Beam Penetration method



- Used with random-scan (vector-scan) monitors.
- Two layers of phosphor, usually red and green, are coated on to the inside of the CRT screen.
- The displayed color depends on how far the electron beam penetrates into the phosphor layers.
- A beam of slow electrons excites only the outer layer i.e. the Red colored layer whereas the beam of very fast electrons penetrates through the red layer and excites the inner green layer.
- At an intermediate speed, the combination of red and green light, i.e. TWO additional colors orange and yellow are generated.

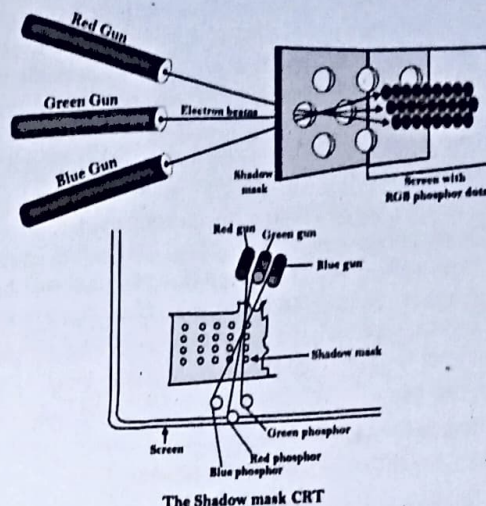
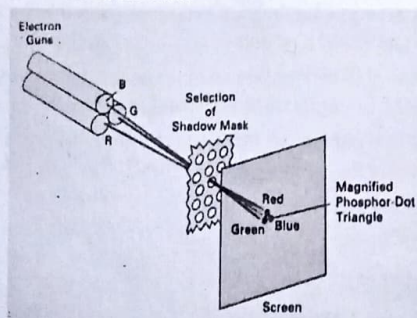
Advantages:

- Half the cost of shadow mask method.
- Provides high resolution of pictures.

Disadvantages:

- The change of color takes time which doesn't suit for interactive graphics at all.
- Less colors are available.
- Quality of picture is not so good (poor).

b. Shadow - Mask method



Workflow

- **Electron Beam Generation:** Inside the CRT, three electron guns emit three electron beams: one for each primary color, red, green, and blue (RGB).
- **Electron Beam Acceleration and Focusing:** The emitted electron beams are accelerated by an anode and then focused by focusing electrodes or magnetic fields. This ensures that each electron beam remains tightly focused and narrow.
- **Shadow Mask Placement:** Positioned just behind the phosphor-coated screen is a shadow mask. This mask is a thin metal sheet with tiny holes or slots that align with the triads of phosphor dots on the screen.
- **Phosphor Triads and Color Mixing:** The screen of a color CRT is coated with phosphor dots arranged in groups of three, forming triads. Each triad consists of red, green, and blue phosphors. The electron beams pass through the holes or slots in the shadow mask and strike the corresponding phosphor dots.
- **Color Selection and Excitation:** As the electron beams hit the phosphor dots, the phosphors are excited and emit light. The intensity of each electron beam is varied to control the amount of light emitted by the corresponding phosphor, thus enabling the selection of specific colors.

- **Color Merging and Image Formation:** The combination of the red, green, and blue light emitted by the excited phosphors creates a full spectrum of colors. By varying the intensities of the electron beams and the phosphor excitation, the CRT generates the desired color for each pixel. The electron beams scan across the screen in a raster pattern, creating the complete color image.
- **Persistence of Vision:** The phosphors on the screen have a short persistence, meaning they continue to emit light for a brief period after being struck by the electron beams. This persistence, coupled with the rapid scanning of the electron beams, creates the illusion of a continuous and stable color image to the human eye.

Advantage:

- Realistic image
- Million different colors to be generated
- Shadow scenes are possible

Disadvantage:

- Relatively expensive compared with the monochrome CRT.
- Relatively poor resolution
- Convergence Problem

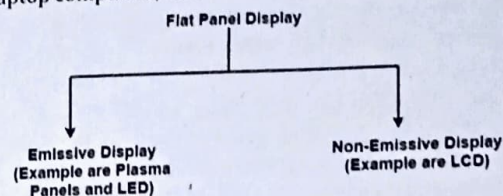
3. Flat Panel Display:



- The Flat-Panel display refers to a class of video devices that have reduced volume, weight and power requirement compare to CRT.
- Flat panel displays are thin, lightweight electronic visual display devices that use flat-panel technology to present visual information.
- They are commonly found in devices such as laptops, computer monitors, televisions, smartphones, and tablets.

- Flat panel displays utilize various technologies such as LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode), and LED (Light Emitting Diode) to produce images with high clarity, color accuracy, and contrast.
- These displays offer several advantages over traditional CRT displays, including space efficiency, energy efficiency, and reduced eye strain.
- Flat panel displays have a wide range of applications, from personal electronics to professional displays in industries such as gaming, entertainment, healthcare, and signage.
- The advancement of flat panel display technology has led to the development of flexible and curved displays, enabling new possibilities in design and user experience.

Example: Small T.V. monitor, calculator, pocket video games, laptop computers, an advertisement board in elevator.



- Emissive Display:** The emissive displays are devices that convert electrical energy into light. Examples are Plasma Panel, thin film electroluminescent display and LED (Light Emitting Diodes).
- Non-Emissive Display:** The Non-Emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns. Examples are LCD (Liquid Crystal Device).

4. Plasma Panel Display:

- Plasma-Panels are also called as Gas-Discharge Display. It consists of an array of small lights. Lights are fluorescent in nature. Plasma panel displays, also known as plasma screens, are a type of flat panel display technology used to create high-quality visual displays.
- Plasma displays consist of tiny cells filled with ionized gas (plasma) that emit ultraviolet light when electrically stimulated.
- The ultraviolet light then strikes phosphors coating the cells, causing them to emit visible light and create the desired colors.
- Plasma panel displays offer several advantages, including wide viewing angles, high contrast ratios, and excellent color reproduction.

- While plasma panel displays were popular for a period, their usage has declined with the rise of LCD and OLED technologies, which offer similar or better performance with improved energy efficiency.

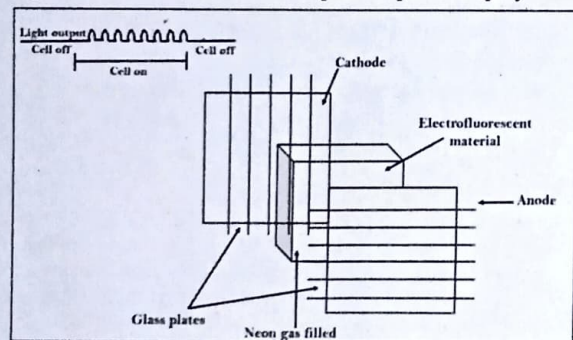
The essential components of the plasma-panel display are:

1. **Cathode:** It consists of fine wires. It delivers negative voltage to gas cells. The voltage is released along with the negative axis.
2. **Anode:** It also consists of line wires. It delivers positive voltage. The voltage is supplied along positive axis.
3. **Fluorescent cells:** It consists of small pockets of gas liquids when the voltage is applied to this liquid (neon gas) it emits light.
4. **Glass Plates:** These plates act as capacitors. The voltage will be applied, the cell will glow continuously.

The gas will glow when there is a significant voltage difference between horizontal and vertical wires. The voltage level is kept between 90 volts to 120 volts. Plasma level does not require refreshing. Erasing is done by reducing the voltage to 90 volts.

Each cell of plasma has two states, so cell is said to be stable. Displayable point in plasma panel is made by the crossing of the horizontal and vertical grid. The resolution of the plasma panel can be up to 512×512 pixels.

Figure shows the state of cell in plasma panel display:



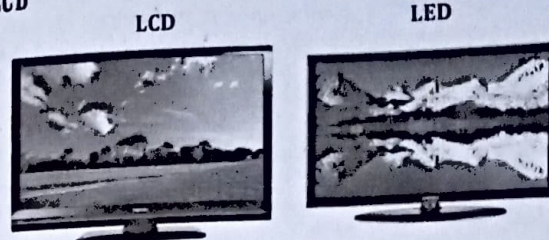
Advantage:

1. High Resolution
2. Large screen size is also possible.
3. Less Volume
4. Less weight
5. Flicker Free Display

Disadvantage:

1. Poor Resolution
2. Wiring requirement anode and the cathode is complex.
3. Its addressing is also complex.

5. LCD



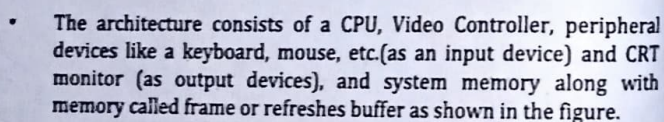
- LCD (Liquid Crystal Display) is a widely used display technology that utilizes liquid crystal molecules to control the passage of light through the screen.
- LCD displays consist of a backlight that illuminates the pixels, a layer of liquid crystals that manipulate the light, and color filters that produce the desired colors.
- LCD displays offer several advantages, including high resolution, sharp image quality, and the ability to display a wide range of colors.
- LCD technology is commonly found in various devices such as computer monitors, televisions, smartphones, tablets, and digital signage.
- LCD displays have evolved over time, with advancements such as LED backlighting, IPS (In-Plane Switching) panels for improved viewing angles, and higher refresh rates for smoother motion.

6. LED (non emissive)

- LED (Light Emitting Diode) is a popular display technology that uses tiny light-emitting diodes to produce light and create visual displays.
- LED displays offer several advantages, including high brightness levels, energy efficiency, and a wide color gamut for vibrant and accurate color reproduction.
- LED technology allows for thinner and more lightweight displays compared to traditional display technologies like CRT or plasma.

- ## 2.3 Raster and Random Systems and Architectures

Raster Scan Architecture



-
- ```

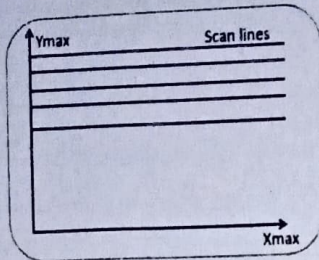
graph TD
 CPU[CPU] <--> IO[I/O port]
 IO -- "(Display commands)" --> DC[Display controller]
 KM[Keyboard/Mouse] -- "(interaction data)" --> DC
 DC --> VC[Video controller]
 VC --> RB[Refresher buffer]
 RB --> Mon[Monitor]

```

### Raster Scan Display

- The frame buffer holds the set of intensity values for all screen points.
- A video controller, often referred to as a video or graphics card, generates graphic information for any video display device, such as a monitor or projector. They are also known as graphics or video adapters.
- Some modern computers do not include video cards but rather have graphics processing units directly integrated into the computer's motherboard.
- The video controller retrieves the stored intensity values from the frame buffer and displays them on the screen one row at a time.
- Frame buffer location and the corresponding screen points are referenced in Cartesian coordinates.
- Scan lines are then labeled from  $Y_{\max}$  at the top of the screen to 0 at the bottom.
- Along each scan, screen pixel positions are labeled from 0 to  $X_{\max}$ .
- The application programs and graphic subroutine packages reside in the system memory and executes on CPU.

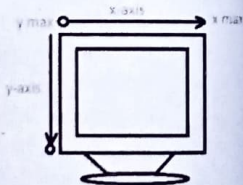
The raster scan proceeds as follows :



- Starting from the top left corner of the screen, the electron gun scans horizontally from left to right one scan line at a time. This is called horizontal retrace.
- Jumping (without tracing the line) to the left end of the next lower row until the bottom right corner is reached.
- Then it jumps (again without tracing) to the top left corner and starts again, finishing one complete refresh cycle. This is called vertical retrace.
- Here, the beam is swept back and forth from left to right across the screen.
- When the beam is moved from the left to the right, it is ON else OFF as shown in the figure.
- Refreshing of the screen is completed at about  $(1/30)$  th of the second.
- In some raster scan systems, each frame is displayed in two passes using an interlaced refresh procedure.
- During this process, instead of refreshing every line of the screen, the electron guns sweep alternate lines of each pass.
- On the first pass, odd-numbered lines are refreshed and in the second pass even numbered lines are refreshed thereby decreasing the time to  $(1/60)$ th of second.

### Video Controller

- In a raster scan system, the video controller plays a significant role in generating and controlling the display of images on the screen. Here's how the video controller functions in a raster scan system:
- **Pixel Generation:** The video controller generates pixels to form an image on the screen. It



receives digital data representing the image from the computer's memory or graphics processing unit (GPU).

- **Horizontal and Vertical Sync:** The video controller produces synchronization signals called horizontal and vertical sync signals. These signals ensure that the electron beam in a cathode ray tube (CRT) or the display mechanism in other types of screens moves systematically across the screen in a coordinated manner.
- **Timing Control:** The video controller controls the timing of the electron beam or display mechanism, ensuring that the screen is painted in a sequential manner from left to right and top to bottom. It manages the timing signals to determine when each pixel should be activated to create the desired image.
- **Frame Buffer:** The video controller typically includes a frame buffer, a dedicated section of memory that stores the pixel data for the entire screen. The frame buffer holds the information required to display each pixel's colour, intensity, and position on the screen.
- **Addressing and Scanning:** The video controller uses a process called scanning or rasterization to traverse the frame buffer and retrieve the pixel data in a sequential manner. It accesses each pixel's colour information from memory and sends it to the display mechanism for rendering on the screen.
- **Colour and Intensity Control:** The video controller can manipulate the colour and intensity of the pixels being displayed. It may apply colour corrections, gamma adjustments, and other image enhancements to ensure accurate and vibrant colour reproduction.
- **Pixel Clock:** The video controller generates a pixel clock signal, which determines the rate at which pixels are displayed on the screen. The pixel clock synchronizes the speed of pixel generation with the display's refresh rate, ensuring a smooth and flicker-free image.
- **Output Interface:** The video controller connects to the display device through an appropriate output interface, such as VGA, DVI, HDMI, or DisplayPort. It converts the digital pixel data into a format compatible with the specific display interface.
- **Resolution and Refresh Rate:** The video controller controls the resolution and refresh rate of the display. It determines the number of pixels horizontally and vertically on the screen and the rate at which the screen is refreshed with new images. Higher resolutions and refresh rates require more processing power and memory bandwidth.
- **Graphics Acceleration:** In addition to its primary functions, some video controllers may include graphics acceleration capabilities.

These features offload complex rendering tasks from the CPU to specialized circuitry or dedicated GPUs, resulting in improved performance and graphical effects.

- Overall, the video controller in a raster scan system orchestrates the display process by generating pixels, synchronizing the electron beam or display mechanism, accessing pixel data from memory, and controlling the colour, intensity, and timing of the displayed image.

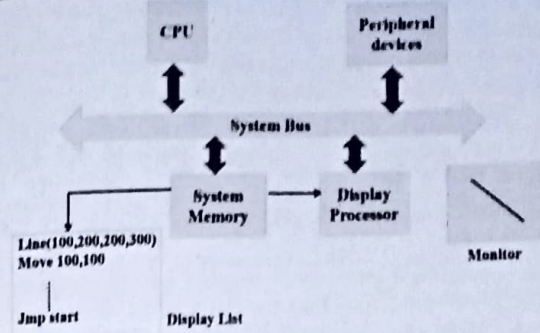
#### Advantages of Raster Scan Architecture:

- **Simplicity:** Raster scan architecture is relatively simple and easy to implement, making it cost-effective for display devices.
- **Compatibility:** Raster scan displays are compatible with most software and content that is designed for raster-based systems.
- **Sharpness and Clarity:** Raster scan displays can achieve high-resolution images with sharp details and clarity, especially in stationary or slow-moving images.
- **Smooth Motion:** Raster scan architecture allows for smooth motion rendering, as each pixel is individually refreshed in a sequential manner.
- **Cost-Effectiveness:** Raster scan displays tend to be more affordable compared to other display technologies, making them accessible to a wide range of users.

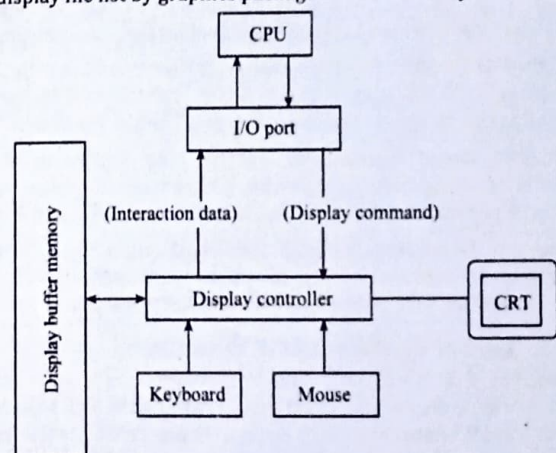
#### Disadvantages of Raster Scan Architecture:

- **Motion Artifacts:** Rapidly moving objects or fast-paced motion on a raster scan display can result in motion artifacts, such as blurring or ghosting.
- **Limited Refresh Rate:** Raster scan displays have a fixed refresh rate, which can lead to perceived flickering, especially in lower refresh rate displays.
- **Limited Viewing Angles:** Raster scan displays often have narrower viewing angles, meaning that the image quality may degrade when viewed from off-center positions.
- **Interlacing Issues:** Some raster scan displays utilize interlaced scanning, which can introduce flickering and reduced vertical resolution in certain scenarios.
- **Inefficient for Graphics and Animation:** Raster scan architecture may not be the most efficient for rendering complex graphics or fast-moving animations, as it requires refreshing the entire screen with each frame.

#### Random or Vector Scan Architecture



- The architecture of vector display consists of a CPU, a display processor, input devices like monitor and output devices like CRT monitor, and system memory along with display list as shown in the figure.
- An application program is input and stored in the system memory along with a graphics package and they get executed in the CPU.
- Graphics commands in the application program are translated to display file list by graphics package and stored in system memory.



Random scan display

- This display file list is accessed by the display processor to refresh the screen.

- The display processor in a random scan system is referred to as a display processing unit or graphics controller.
- Instead of containing mapped pixel values as in the raster system, the display list consists of a set of line drawing commands or other commands related to picture generations.
- The DPU executes each command from the display file list during every refresh cycle and feeds the vector generator with digital x and y co-ordinated.
- The vector generator converts the digital signals into equivalent analog voltage.
- This causes the electron beam to move directly to the start point towards the end point. Thus the beam swept does not follow any fixed pattern, the direction is arbitrary as directed by the display commands.

#### Advantages of Random Scan Architecture:

- **Efficient for Graphics and Animation:** Random scan architecture is specifically designed for rendering graphics and animations efficiently, making it suitable for applications that require fast and dynamic image generation.
- **Smooth and Accurate Motion:** Random scan displays can provide smooth and accurate motion rendering, as they have the capability to directly address and update individual pixels or image elements.
- **High Resolution:** Random scan architecture allows for high-resolution displays, as it can precisely control the position and intensity of each pixel, resulting in detailed and sharp images.
- **Flexible Image Generation:** Random scan displays offer more flexibility in generating complex images, as they can directly manipulate individual pixels, enabling sophisticated visual effects.
- **Reduced Flickering:** Random scan architecture typically utilizes higher refresh rates, reducing the perceived flickering on the screen and providing a more comfortable viewing experience.

#### Disadvantages of Random Scan Architecture:

- **Complexity and Cost:** Random scan architecture is more complex and costly to implement compared to raster scan architecture, as it requires additional hardware and processing capabilities to address and update individual pixels.
- **Limited Compatibility:** Random scan displays may have limited compatibility with software and content designed for raster-based systems, requiring additional software adaptations or specialized programming.

- **Lower Refresh Rates:** Achieving high refresh rates in random scan displays can be challenging, resulting in potential limitations in fast-paced or high-motion scenarios.
- **Limited Availability:** Random scan architecture is less common in consumer display devices, as it has been largely replaced by more cost-effective and efficient technologies like raster-based displays, LCD, OLED, and LED.
- **Viewing Angle Issues:** Some random scan displays may exhibit viewing angle limitations, where the image quality or color accuracy degrades when viewed from certain angles.
- Overall, random scan architecture offers advantages in terms of graphics rendering, motion

#### Raster vs Random

|                         | Raster Scan                                                                    | Random Scan                                                                       |
|-------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Display Technique       | Scans the screen in a systematic pattern from top to bottom                    | Scans the screen randomly based on the content                                    |
| Image Generation        | Renders images line by line in a sequential manner                             | Renders images by directly addressing and updating individual pixels              |
| Motion Rendering        | Can result in motion artifacts or blurring for fast-moving objects             | Offers smoother and more accurate motion rendering                                |
| Refresh Rate            | Fixed refresh rate for the entire screen 60-80 fps                             | Variable refresh rate based on the content being displayed 30 to 60 times per sec |
| Efficiency for Graphics | Less efficient for complex graphics or animations                              | More efficient for graphics and animation rendering                               |
| Display Complexity      | Relatively simpler to implement                                                | More complex and requires additional hardware                                     |
| Compatibility           | Compatible with most software and content designed for raster-based systems    | May require additional software adaptations or specialized programming            |
| Resolution              | Resolution determined by the number of pixels per line and the number of lines | Resolution determined by the display's capabilities to address individual pixels  |

|              | Raster Scan                                                                   | Random Scan                                                                                    |
|--------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Flickering   | Can have perceived flickering, especially at lower refresh rates              | Higher refresh rates reduce perceived flickering                                               |
| Smoothness   | Produce jagged lines                                                          | Produce smooth lines                                                                           |
| Editing      | Difficult                                                                     | Easy                                                                                           |
| Cost         | Cheaper                                                                       | expensive                                                                                      |
| Interlacing  | It uses interlacing                                                           | It doesn't use                                                                                 |
| Availability | Widely used in older CRT displays, less common in modern display technologies | Less common in consumer display devices, replaced by raster-based displays, LCD, OLED, and LED |
| Example      | CRT, TV, Printer                                                              | Pen, Plotter                                                                                   |

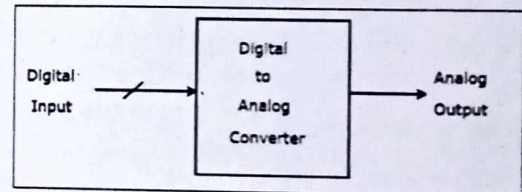
#### LCD vs LED

|                   | LCD Displays                                              | LED Displays                                                           |
|-------------------|-----------------------------------------------------------|------------------------------------------------------------------------|
| Technology        | Uses liquid crystal molecules                             | Uses light-emitting diodes (LEDs)                                      |
| Light Source      | Requires a separate backlight                             | Utilizes LEDs as the light source                                      |
| Energy Efficiency | Consumes more power                                       | Energy-efficient and consumes less power                               |
| Thickness         | Tends to be thicker                                       | Thinner and more compact                                               |
| Contrast Ratio    | Lower contrast ratio                                      | Higher contrast ratio                                                  |
| Color Accuracy    | Slightly less vibrant colors                              | Improved color accuracy and saturation                                 |
| Response Time     | Slower response time                                      | Faster response time, better for fast-paced content                    |
| Viewing Angles    | Limited viewing angles, color distortion from side angles | Wide viewing angles, consistent color reproduction from various angles |
| Lifespan          | Typically has a longer lifespan                           | LEDs have a longer lifespan                                            |
| Cost              | Generally more affordable                                 | Initially higher cost, but prices have reduced over time               |

#### Interlaced vs Progressive Scan

| S.NO | Interlaced Scan                                                                 | Progressive Scan                                                                       |
|------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1.   | In interlaced scan, scanning takes place over dividing one frame.               | While in progressive scan, scanning takes place through scanning all frame promptly.   |
| 2.   | Interlaced scan is less efficient than progressive scan.                        | While progressive scan is more efficient than interlaced scan.                         |
| 3.   | In interlaced scan, the displaying video speed is lesser than progressive scan. | While in progressive scan, the displaying video speed is quicker than interlaced scan. |
| 4.   | There is present the combing effect in interlaced scan.                         | While there is not present combing effect in progressive scan.                         |
| 5.   | In interlaced scan, the video quality is vulgarized.                            | While in progressive scan, the video quality is superior than interlaced scan.         |
| 6.   | An interlaced scan is less promoted than a progressive scan.                    | While the progressive scan is more promoted than interlaced scan.                      |

#### Digital to Analog Converter



A digital-to-analog converter (DAC) is a device or circuit that converts digital signals into analog signals. It is commonly used to convert digital data from a computer or digital system into analog signals suitable for various applications, such as audio playback, video output, control systems, and more. Here are some key points about digital-to-analog converters:

- **Purpose:** The primary purpose of a DAC is to transform discrete digital values (typically binary) into continuous analog signals that can represent varying voltage levels or physical quantities.
- **Binary to Analog Conversion:** DACs use various techniques to convert binary data into analog signals. Some common methods

include pulse width modulation (PWM), pulse density modulation (PDM), resistor ladder networks, and sigma-delta modulation.

- **Resolution:** DACs have a specified resolution, which represents the number of discrete levels or steps the converter can produce in the analog output signal. Higher-resolution DACs can provide more accurate and finely graded analog signals.
- **Sampling Rate:** DACs are often designed to work with specific sampling rates, which determine the rate at which the digital input signal is converted into an analog output signal. The sampling rate affects the maximum frequency that can be accurately reproduced by the DAC.

### Solved Numerical Problems

1. A system with 24 bits per pixel and resolution of 1024 by 1024. Calculate the size of frame buffer (in Megabytes)

#### Solution:

$$\text{Resolution} = 1024 * 1024$$

$$\text{Total number of pixel} = 1024 * 1024 = 1048576 \text{ pixels}$$

$$\text{Bits per pixels' storage} = 24 \text{ bits}$$

$$\text{Therefore, total storage required in frame buffer:}$$

$$= 1048576 * 24$$

$$= 25165824 \text{ bits}$$

$$= 25165824 / 8 \text{ Byte}$$

$$= 25165824 / (8 * 1024) \text{ Kb}$$

$$= 25165824 / (8 * 1024 * 1024) \text{ Mb}$$

$$= 3 \text{ Mb}$$

2. How Many k bytes does a frame buffer needs in a 600 x 400 pixel?

#### Solution:

Suppose, n bits are required to store 1 pixel.

Then, the size of frame buffer = Resolution \* bits per pixel

$$= (600 * 400) * n \text{ bits}$$

$$= 240000 n \text{ bits}$$

$$= 240000 n \text{ kb} / (8 * 1024)$$

$$= 29.30 n \text{ k bytes.}$$

3. Consider a RGB raster system is to be designed using 8 inch by 10-inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer, how much storage in bytes do we need for the frame buffer?

#### Solution:

$$\text{Size of screen} = 8 \text{ inch} * 10 \text{ inch}$$

$$\text{Pixels Per inch (Resolution)} = 100$$

$$\text{Then total number of pixels} = 8 * 100 * 10 * 100$$

$$= 800000 \text{ pixels}$$

$$\text{Bits per pixels' storage} = 8$$

$$\text{Therefore, total storage required in frame buffer}$$

$$= 800000 * 8 \text{ bits}$$

$$= 6400000 \text{ bits}$$

$$= 6400000 / 8 \text{ bytes}$$

$$= 800000 \text{ Bytes}$$

4. Find out the aspect ratio of the raster system using 8 x 10 inches' screen and 100 pixels/inch.

#### Solution:

We know that, Aspect ratio = Width / Height

$$= (8 * 100) / (10 * 100)$$

$$= 4 / 5$$

$$\text{So, aspect ratio} = 4: 5$$

5. What is the time required to display a pixel on the monitor of size 1024 x 768 with refresh rate of 60 Hz?

#### Solution:

Refresh Rate = 60Hz i.e. 60 frames per second

$$\text{Total number of pixel in one frame} = 1024 * 768 = 786432 \text{ pixels.}$$

60 frames need 1 second

$$1 \text{ frame need } \frac{1}{60} \text{ second}$$

$$\text{i.e. } 786432 \text{ pixels need } \frac{1}{60} \text{ second}$$

$$\text{Hence, 1 pixel need } \frac{1}{60 * 786432} \text{ second}$$

$$= \frac{10^9}{60 * 786432} \text{ ns} = 21.19 \text{ ns}$$

6. If the total intensity achievable for a pixel is 256 and the screen resolution is  $640 \times 480$ . What will be the size of frame buffer?

**Solution:**

1 pixel = 256 different intensity level

Resolution =  $640 \times 480$

Let X be the number of bits required to represent 256 different intensity level

Then,  $2^x = 256$

Therefore,  $x = 8$  bits

Hence, number of bits required for the screen =  $640 \times 480 \times 8$   
= 2457600 bits.

7. If a pixel is accessed from the frame buffer with an average access time of 300ns then will this rate produce an unflickering effect for the screen size of  $640 \times 480$ .

**Solution:**

Size of screen =  $640 \times 480$

Total Number of pixels =  $640 \times 480 = 307200$

Average access time of one pixel = 300ns

Therefore, total time required to access entire pixels of image in the screen =  $307200 \times 300$

$$= 92160000 \text{ ns}$$

$$= 92160000/10^9$$

$$= 0.09216 \text{ seconds}$$

i.e. 1 cycle take 0.09216 second

Now, number of cycles per second i.e. Refresh Rate = ?

$$0.09216 \text{ Seconds} = 1 \text{ cycle}$$

$$1 \text{ second} = \frac{1}{0.09216}$$

$$= 10.86$$

Therefore, refresh Rate = 10.86 cycles per second

Since the minimum refresh rate for unflicker image is 60 frames per second, hence we can say the monitor produces flickering effect.

8. Consider a raster scan system having 12-inch by 10-inch screen with a resolution of 100 pixels per inch in each direction. If display controller of this system refresh the screen at the rate of 50 frames per second. How many pixels could be accessed per second and what is the access time per pixel of the system. (TU Exam)

**Solution:**

Given,

Resolution =  $12 \times 10$  (inch)

Pixel per inch = 100 ppi

Refresh rate = 50 fps

Total access time for a pixel = ?

Total pixels accessed in a sec = ?

Now,

$$\text{Total no. of pixels in 1 frame} = 12 \times 100 \times 10 \times 100 \\ = 1200000$$

$$\text{Total no. of pixels in 50 frame} = 1200000 \times 50 \\ = 60000000$$

50 frame needs 1 sec.

60000000 pixels needs 1 sec

$$\therefore 1 \text{ pixels needs } \frac{1}{60000000} \text{ sec}$$

$$= \frac{1.666 \times 10^{-8}}{10^{-9}}$$

$$= 16.67 /s$$

9. Consider three different raster systems with resolutions of  $640 \times 480$ ,  $1280 \times 1024$  and  $2560 \times 2048$ . What size of frame buffer (in bytes) is needed for each of these systems to store 12 bits per pixel? How much storage is required for each system if 24 bits per pixel are to be stored? (UPTU MCA 2003)

**Solution:**

Case 1: For  $640 \times 480$ ,

$$\text{Total number of pixel required for } 640 \times 480 \text{ resolution} \\ = 640 \times 480 \text{ pixels}$$

So, size of frame-buffer required =  $640 \times 480$  pixels  
because 1 pixel can store 12 bits.

Therefore, size of frame buffer (in bits)

$$= 640 \times 480 \times 12 \text{ bit}$$

$$= \frac{640 \times 480 \times 12}{8} \text{ bytes (1 byte = 8 bit)}$$

$$= 460800 \text{ bytes}$$

$$= 450 \text{ KB (1024 bytes = 1 KB)}$$

For 1280 × 1024: Size of frame buffer (in bits)

$$= 1280 \times 1024 \times 12 \text{ bits}$$

$$= \frac{1280 \times 1024 \times 12}{8} \text{ bytes}$$

$$= 1920 \text{ KB}$$

For 2560 × 2048:

$$\text{Size of frame-buffer} = \frac{256 \times 2048 \times 12}{8} \text{ bytes} = 7.5 \text{ MB}$$

Case II: When one pixel can store 24 bits then frame-buffer size will be twice.

$$\text{For } 640 \times 480, \text{ Frame-Buffer size} = \frac{640 \times 480 \times 24}{8} \text{ byte}$$

$$= 900 \text{ KB}$$

$$\text{For } 1280 \times 1024, \text{ Frame-buffer size} = 1920 \times 2 \text{ KB}$$

$$= 3840 \text{ KB} = 3.75 \text{ MB}$$

$$\text{For } 2560 \times 2048, \text{ Frame-buffer size} = 7.5 \times 2 \text{ MB} = 15 \text{ MB.}$$

10. Consider a raster system with a resolution of 1024 × 1024. What is the size of the raster (in bytes) needed to store 4 bits per pixel? How much storage is required if 8 bits per pixel are to be stored? (UPTU, B-Tech, 2002)

Solution:

$$\text{Resolution} = 1024 \times 1024$$

Case-1: When 1 pixel = 4 bits

$$\text{Size of raster} = 1024 \times 1024 \times 4 \text{ bits}$$

$$= \frac{1024 \times 1024 \times 4}{8} \text{ bytes (since, 1 byte = 8 bits)}$$

$$= 515 \times 1024 = 1 \times 2^{19} \text{ bytes}$$

Case-2: When 1 pixel = 8 bits

$$\text{Size of raster} = 1024 \times 1024 \times 8 \text{ bits}$$

$$= \frac{1024 \times 1024 \times 8}{8} \text{ bytes}$$

$$= 1 \times 2^{20} \text{ bytes}$$

11. Consider two raster systems with resolution of 640 by 480 and 1280 by 1024. How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second? What is the access time per pixel in each system?

Solution:

Case-1: Resolution = 640 × 480

$$\text{No. of pixels in one frame} = 690 \times 480 = 307200$$

Since controller can access 60 frames in one second

$$\text{Therefore, total no. of pixels accessed} = 60 \times 307200$$

$$= 18432000 \text{ per sec}$$

$$\text{Access time/pixel} = \frac{1}{\text{Total pixels accessed per sec}}$$

$$= 5.4 \times 10^{-8} \text{ sec/pixel.}$$

Case-2: When resolution = 1280 × 1024

$$\text{Total no. of pixels accessed by the raster system}$$

$$= 60 \times 1280 \times 1024$$

$$= 78643200 \text{ per sec.}$$

$$\text{Access time per pixel} = 1178643200 = 1.2 \times 10^{-8} \text{ sec/pixel}$$

12. A laser printer is capable of printing two pages (size 9 × 11 inch) per second at resolution of 600 pixels per inch. How many bits per second does such device require?

Solution:

Storage required per page (in pixel)

$$= 9 \times 11 \text{ inch/sec}$$

$$= 9 \times 11 \times 600 \times 600 \text{ pixel sec } (\because 1 \text{ inch} = 600 \text{ pixels})$$

Since laser printer is capable printing two pages per second bits per second required by the printer.

$$= 2 \times 9 \times 11 \times 600 \times 600 \text{ pixels per second}$$

$$= 7.128 \times 10^7 \text{ pixel/sec}$$

$$= 7.128 \times 10^7 \text{ n bits/sec}$$

$$(\text{Assume } 1 \text{ pixel} = \text{n bits})$$

13. How many k bytes does a frame buffer need in a 800 × 400 pixel?

Solution:

Given resolution is 600 × 400

Suppose 1 pixel can store n bits then the size of frame buffer

$$= \text{Resolution} \times \text{bits per pixel}$$

$$= (600 \times 400) \times \text{n bits}$$

$$= 240000 \text{ n bits}$$

$$= \frac{240000 \text{ n}}{1024 \times 8} \text{ k bits } (\because 1 \text{ kb} = 1024 \text{ bits})$$

$$= 29.30 \text{ n k bytes.}$$

14. Find out the aspect ratio of the raster system using  $8 \times 10$  inches screen and 100 pixel/inch.

**Solution:**

We know that,

$$\text{Aspect ratio} = \frac{\text{width}}{\text{height}} = \frac{8 \times 100}{10 \times 100} = \frac{4}{5}$$

$\therefore$  Aspect ratio = 4 : 5

15. How much time is spent scanning across each row of pixels during screen refresh on a raster system with resolution of  $1280 \times 1024$  and a refresh rate of 60 frames per second?

**Solution:**

Here, resolution,  $1280 \times 1024$

That means system contains 1024 scan lines and each scan line contains of 1280 pixels and refresh rate = 60 frames/sec.

that means 1 frame takes  $\frac{1}{60}$  sec

Since, resolution =  $1280 \times 1024$

$\therefore$  1 frame consists of 1024 scan lines.

$\therefore$  1024 scan lines takes to  $\frac{1}{60}$  sec

$\therefore$  1 scan lines take  $\frac{1}{60 \times 1024}$  sec = 0.058 sec or 58 m sec.

16. Suppose RGB raster system is to be designed using on 8 inch  $\times$  10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for frame buffer?

**Solution:**

Here, resolution = 8 inch  $\times$  10 inch

First we convert it in pixel, then

resolution =  $8 \times 100$  by  $10 \times 100$  pixel =  $800 \times 1000$  pixel

1 pixel can store 6 bits

so, frame buffer size required =  $800 \times 1000 \times 6$  bits

$$= \frac{800 \times 1000 \times 6}{8} \text{ bytes}$$

$$= 6 \times 10^5 \text{ bytes}$$

17. How long would it take to load a  $640 \times 480$  frame buffer with 12 bits per pixel, if  $10^5$  bits can be transferred per second? How long would it take to load a 24 bits per pixel frame buffer with a resolution of  $1280$  by  $1024$  using this same transfer rate?

**Solution:**

Frame buffer size =  $640 \times 480$  pixels

$$= 640 \times 480 \times 12 \text{ bits [1 pixel} = 12 \text{ bits]}$$

Since  $10^5$  bits takes 1 second

$$\text{So, } 640 \times 480 \times 12 \text{ bits takes } \frac{640 \times 480 \times 12}{10^5} = 36.864 \text{ sec}$$

18. Consider a raster scan system having 12 inch by 10 inch screen with a resolution of 100 pixels per inch in each direction. If the display controller of this system refreshes the screen at the rate of 50 frames per second, how many pixels could be accessed per second and what is the access time per pixel of the system? [2071 Shravan]

**Solution:**

Total pixels =  $12 \times 100 \times 10 \times 100$

Refresh rate = 50 frames per second

$$\text{Pixels accessed per second (f)} = 12 \times 100 \times 10 \times 100 \times 50 = 60000000$$

$$\text{Access time per pixel} = \frac{1}{f}$$

$$= 1.667 \times 10^{-8} \text{ second}$$

Access time per pixel is  $1.667 \times 10^{-8}$  second

19. Calculate the frame buffer size (in KB) for a raster system recording a video for 1 min with resolution of  $1280 \times 1024$ , and storing 24 bits per pixel with a refresh rate of 25 fps. [2076 Ashwin]

**Solution:**

Screen resolution =  $1280 \times 1024$

Refresh rate = 25 fps

Bit required to represent a pixel = 24 bits

Memory required just for a frame =  $1280 \times 1024 \times 24$  bits

Memory required for 1 second =  $1280 \times 1024 \times 24 \times 25$  bits

Memory required for recording a video for 1 min is

$$= 1280 \times 1024 \times 25 \times 24 \times 60 \text{ bits}$$

$$= 5760,0000 \text{ KB}$$

20. Calculate the total memory required to store a 10 minutes video in a SVGA system with 24 bit true color and 60 fps refresh rate  
[2078 Kartik]

Solution:

The resolution of SVGA system =  $800 \times 600$

Bit required to represent a pixel = 24 bit

Refresh rate = 60 fps

Memory required for one frame =  $800 \times 600 \times 24$

Memory required for 1 second =  $800 \times 600 \times 24 \times 60$  bit

Memory required for recording a video for 10 min is

$$= 800 \times 600 \times 24 \times 60 \times 10 \times 60 \text{ bit}$$

$$= 50,625,000 \text{ KB}$$

21. Let the resolution of screen is  $1024 \times 512$ . What is the memory captured by the frame buffer that uses primary color for display?  
[2078 Chaitra]

Solution:

The resolution of screen =  $1024 \times 512$

For primary color to display, let the color depth = 24 bit

Size of frame buffer for 1 pixel = 24 bit

$$\text{For } 1024 \times 512 \text{ pixels, size of frame buffer} = 1024 \times 512 \times 24 \text{ bit} \\ = 4 \text{ MB}$$

The memory captured by the frame buffer is 4 MB

22. What is the fraction of the total refresh time per frame spent in retrace of the electron beam for a noninterlaced raster system with a resolution of 1280 by 1024, a refresh rate of 60 Hz, a horizontal retrace time of 5 microseconds, and a vertical retrace time of 500 microseconds?

Solution:

$$1 \text{ sec} = 10^6 \text{ usec}$$

$$\text{Refresh rate} = 60 \text{ Hz} = \frac{1}{60} \text{ sec to scan} = 16.7 \text{ msec}$$

$$\text{The time for horizontal retrace} = 1024 \times 5 \text{ usec}$$

$$\text{The time for vertical retrace} = 500 \text{ usec}$$

$$\text{Total time spent for retrace} = 5120 + 500 \\ = 5620 \text{ usec} = 5.62 \text{ msec}$$

The fraction of the total refresh time frame spent in retrace

$$= \frac{5.62}{16.7} = 0.337$$